

Communication

Date	03 July 2026
Team ID	N/A
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Communication Plan:

OptiCrop was developed by a single person (no assigned team), so 'communication' here reflects how progress was tracked and how the work will be communicated to evaluators/mentors, rather than coordination between multiple teammates.

S. No	Communication Type	Frequency	Channel / Tool	Participants	Purpose
1	Team Standup	Daily	Personal progress notes / task list	Solo developer	Track daily progress and identify blockers
2	Progress Update	Weekly	Git commit history and README updates	Solo developer (and mentor, if assigned)	Summarize work completed and next steps
3	Issue / Bug Discussion	As Needed	Debugging notes; GitHub Issues (if used)	Solo developer	Document and resolve technical issues as they arise
4	Stakeholder Review	Bi-Weekly	Optional check-in with mentor/instructor	Solo developer & mentor	Get feedback on project direction and scope
5	Final Demo Rehearsal	Once	Local run-through of the Find Your Crop demo flow	Solo developer	Ensure a smooth, well-timed live demonstration

Communication Challenges & Resolutions:

S.No	Challenge Faced	Resolution / Action Taken
1	Determining realistic soil/climate value ranges for the training dataset	Researched typical agronomic ranges per crop and validated the resulting distributions through EDA (histograms, boxplots)
2	Balancing model accuracy with simplicity for a clear demo	Compared 4 algorithms on standard metrics (accuracy, precision, recall, F1) and selected Random Forest based on the results
3	Realizing the Flask dev server's single-threaded limitation only during performance testing	Documented it as a known limitation and noted Unicorn/WSGI deployment as the recommended fix for production use